BIG DATA MANAGEMENT | SEMESTER VII

QuakeWatch

Spatial and temporal analysis of global earthquakes

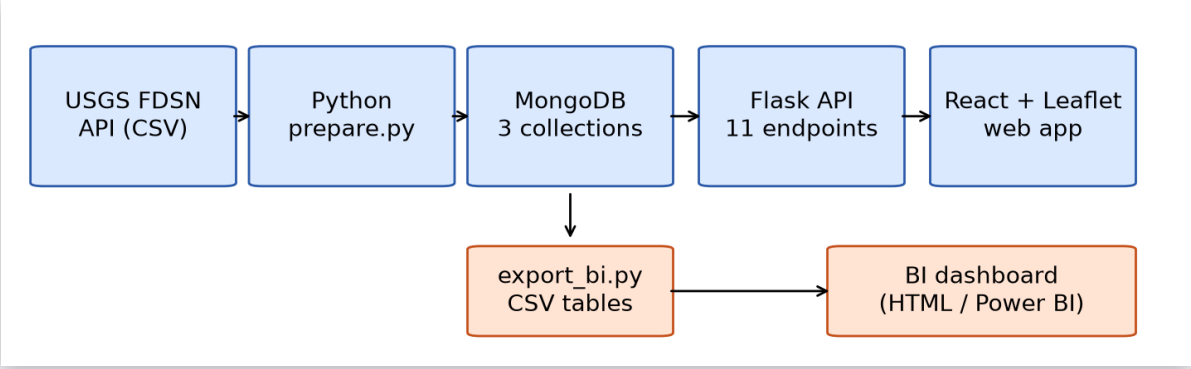
USGS ComCat · MongoDB · Flask · React · BI dashboard

Anurag 2023IMG-008 · Kushagra 2023IMG-030 · Kshitij 2023IMG-029 · Vibhor 2023IMG-054
ABV-IIITM Gwalior · IMG (Integrated B.Tech + MBA) · November 2026

Five questions drive the project

1. **When do earthquakes occur?**
   Volume by month, hour of day and weekday
2. **Which conditions have more strong events?**
   Depth, magnitude type, network, region
3. **Where do events concentrate?**
   Counts versus strong-event concentration
4. **What do indexes save?**
   Including a geospatial 2dsphere query
5. **One dataset, two front ends?**
   Web app and BI dashboard from the same data

From USGS API to app and dashboard



54,394
raw USGS rows, 24 files

53,952
earthquakes kept

55,152
MongoDB documents

11
Flask API endpoints

442 non-earthquake events (mostly mining explosions) removed; 0 duplicates, 0 bad coordinates.

Headline numbers

53,952
earthquakes M2.5+

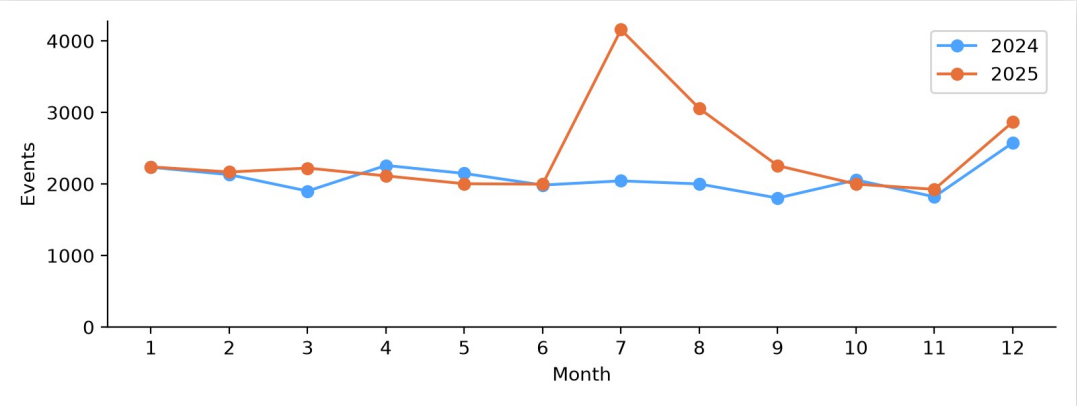
3,636
strong events, M ≥ 5.0

6.74%
strong share

M8.8
Kamchatka, 29 Jul 2025

- 2024: 24,954 events, 6.04% strong, largest M7.5
- 2025: 28,998 events, 7.34% strong, largest M8.8
- M6+ events rose from 99 to 145; M7+ from 10 to 16
- Counts fall about ten-fold per magnitude unit (b ≈ 1.2)

One earthquake reshaped 2025



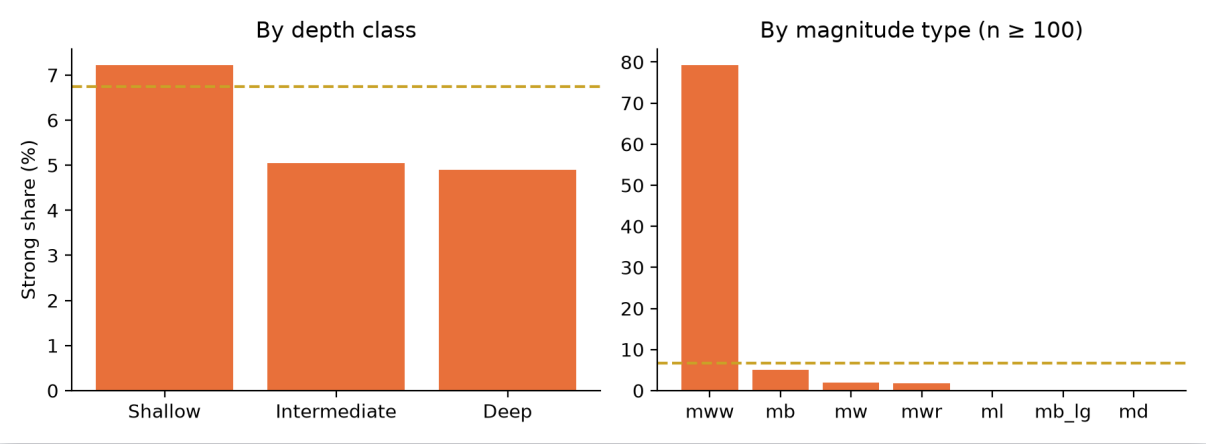
4,157
events in Jul 2025 vs 2,043

82.6%
of 2025 radiated energy

14×
2025 energy vs 2024

Excluding the Kamchatka window, 2025 strong share falls from 7.34% to 6.84%.

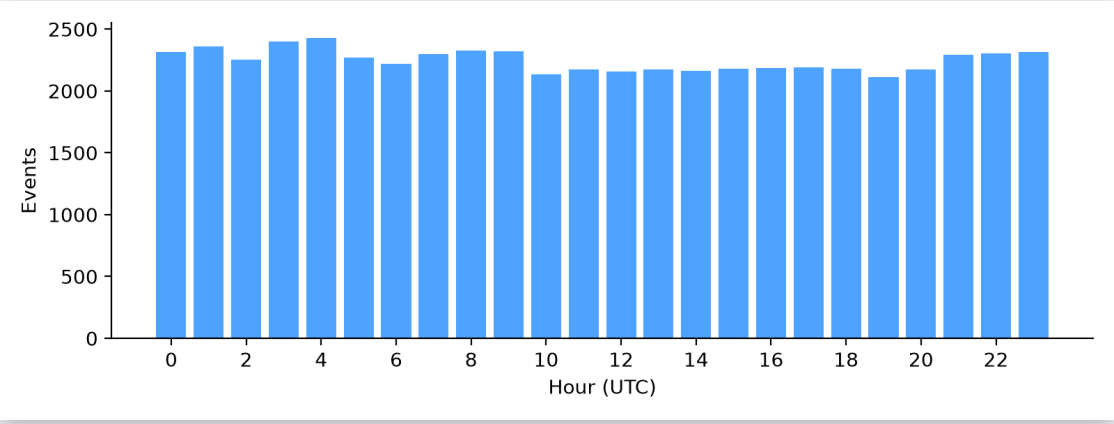
Strong share reflects reporting



- mww 79.3% vs ml 0.07%: moment magnitudes exist mainly for larger events
- US network 8.64% strong; Alaska 0.35%, Puerto Rico 0.16%
- Like for like (mb): shallow 6.39% vs deep 1.46%, a ratio of 4.4

Compare within one magnitude type and contributor, or use M ≥ 4.5 where the catalogue is complete.

Earthquakes keep no clock



2,115–2,431

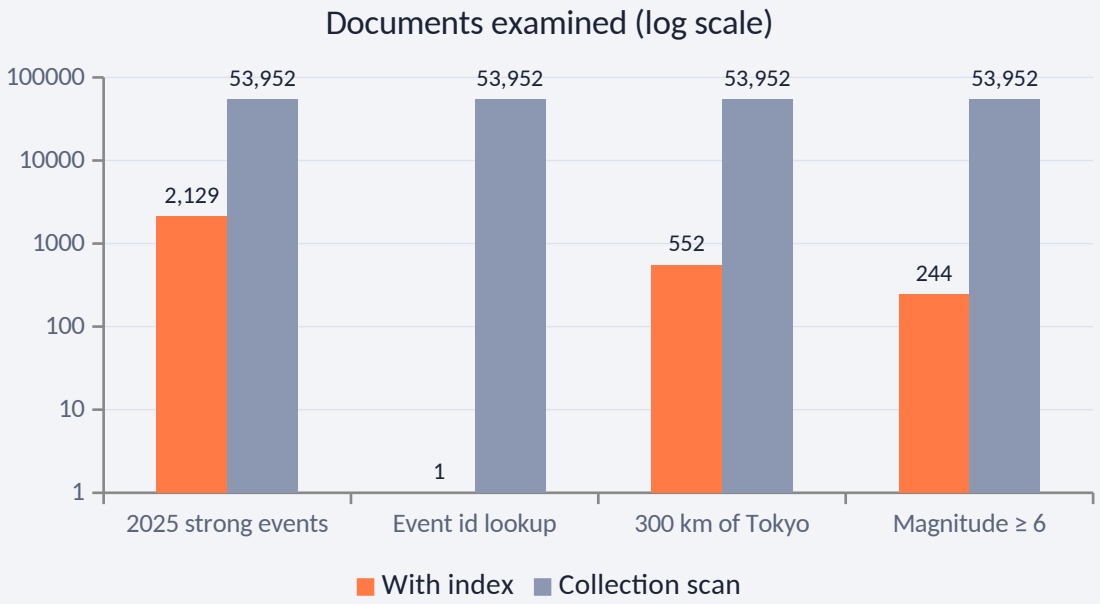
events per hour, flat

7,552–8,028

events per weekday

A flat profile is the expected result and confirms the pipeline's time handling.

Indexes: 25× to 54,000× less work



0 ms

indexed (below timer resolution)

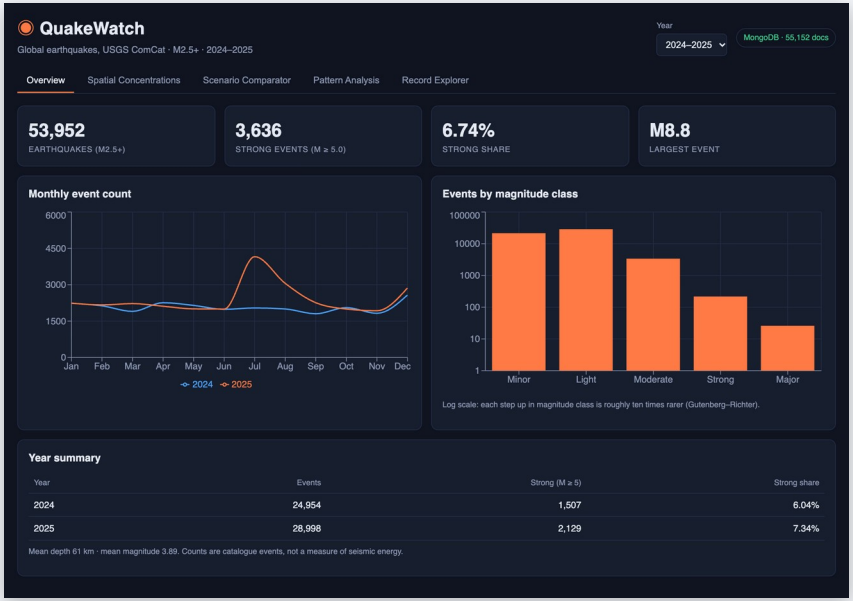
12–41 ms

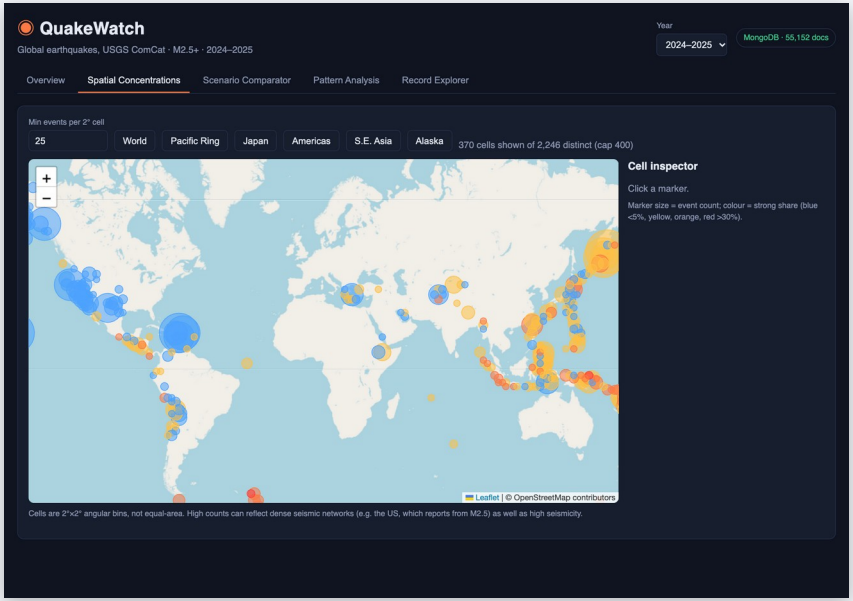
collection scan, 5-run median

MongoDB 8.0.34, 53,952 docs, explain(executionStats). The 2dsphere index with a bounded $geoWithin query is selective: 570 keys for 427 matches.

Document-count ratios transfer to larger data; absolute times at this size do not.

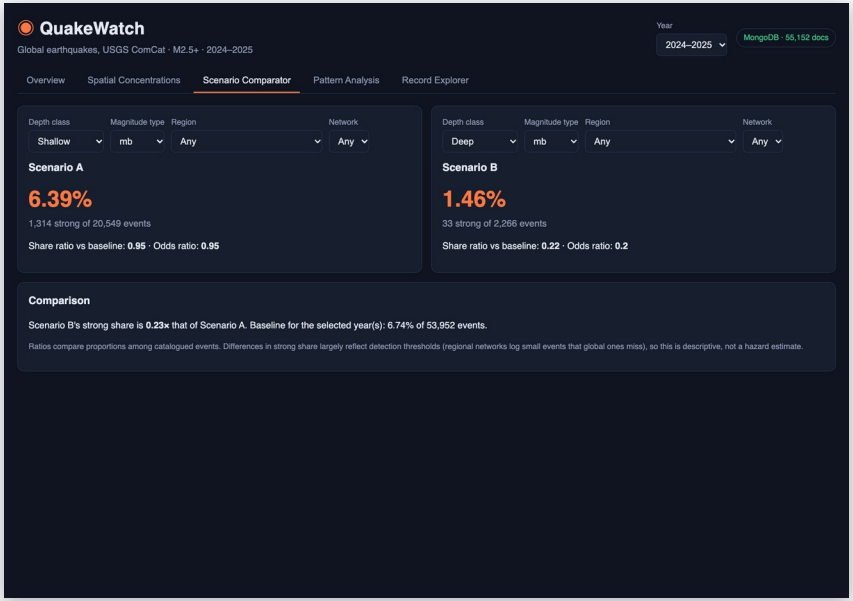
Web app: overview and map

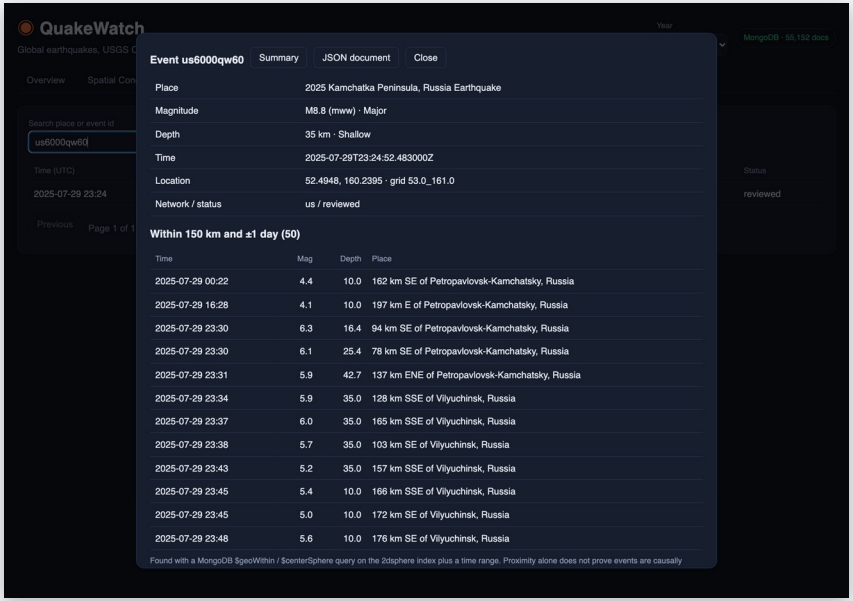




React + TypeScript + Leaflet over a Flask API; five screens, one year control.

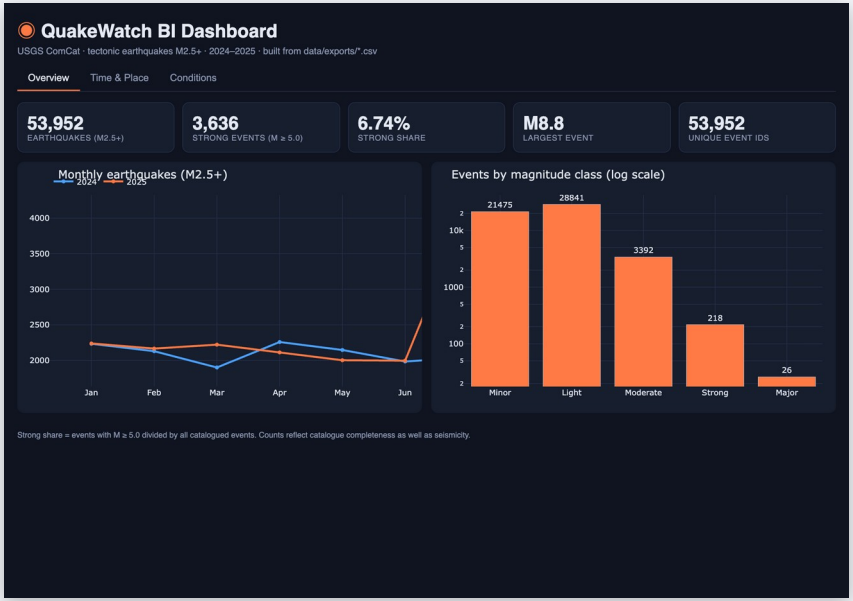
Compare scenarios, inspect events

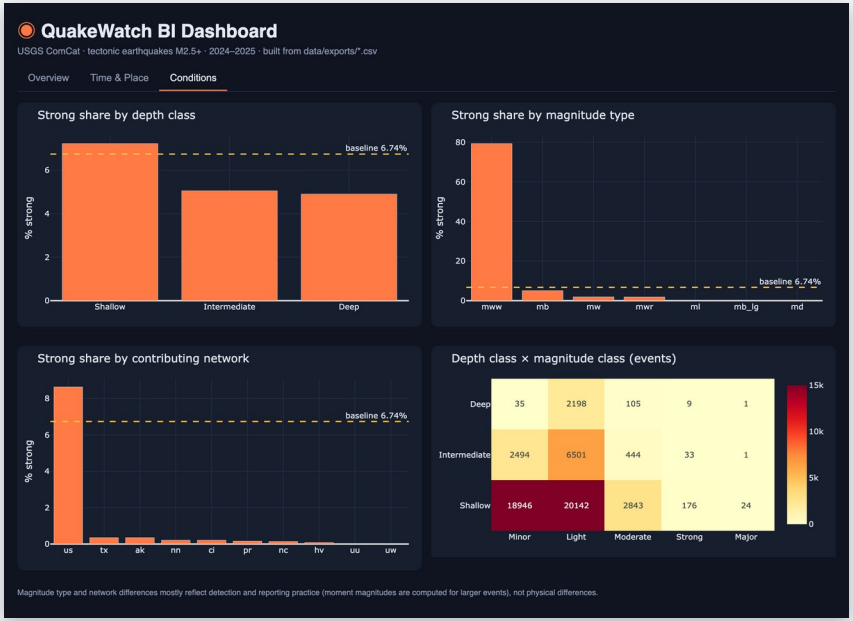




Share and odds ratios against the baseline; $geoWithin finds events within 150 km and one day.

BI dashboard from the same data





Star-schema CSVs plus DAX measures for Power BI; bundled HTML build mirrors the three pages.

What the findings do not show

Catalogue completeness

Networks detect different minimum magnitudes, so counts mix reporting and seismicity.

Two years only

One M8.8 sequence dominates 2025; no trend can be claimed.

Heuristics

Regions parsed from text; 2° cells are not equal-area; no completeness or hazard model.

Dashboard format

HTML build shipped; the Power BI file is assembled from the provided tables and DAX.

Conclusion

- A public catalogue became a queryable MongoDB database, a web app and a dashboard
- Frequency, magnitude and energy tell different stories
- Reporting practice shapes apparent patterns; compare like with like
- Indexes, including 2dsphere, cut documents examined by 1 to 4 orders of magnitude

Code: github.com/vibhor-5/quakewatch-bdm

Report: github.com/vibhor-5/quakewatch-bdm-report